

Family Knowledge Base and Queries in Prolog

Family information:

John is male.

Paul is male.

Mike is male.

David is male.

Mary is female.

Susan is female.

Lisa is female.

Emma is female.

John is the father of **Paul**.

John is the father of **Susan**.

Paul is the father of **Mike**.

Paul is the father of **Lisa**.

Mike is the father of **Emma**.

Mary is the mother of **Paul**.

Mary is the mother of **Susan**.

Susan is the mother of **Mike**.

Susan is the mother of **Lisa**.

Lisa is the mother of **Emma**.

Prolog Knowledge Base

% Family Knowledge Base

% Facts

male(john).

male(paul).

male(mike).

male(david).

female(mary).

female(susan).

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female(lisa).  
female(emma).
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father(john, paul).  
father(john, susan).  
father(paul, mike).  
father(paul, lisa).  
father(mike, emma).
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mother(mary, paul).  
mother(mary, susan).  
mother(susan, mike).  
mother(susan, lisa).  
mother(lisa, emma).
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% Rules

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parent(X, Y) :- father(X, Y).  
parent(X, Y) :- mother(X, Y).
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grandparent(X, Y) :- parent(X, Z), parent(Z, Y).  
sibling(X, Y) :- parent(Z, X), parent(Z, Y), X \= Y.  
brother(X, Y) :- sibling(X, Y), male(X).  
sister(X, Y) :- sibling(X, Y), female(X).  
ancestor(X, Y) :- parent(X, Y).  
ancestor(X, Y) :- parent(X, Z), ancestor(Z, Y).
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Queries and Answers

1. ?- parent(john, susan).
2. ?- parent(susan, mike).
3. ?- grandparent(john, mike).
4. ?- grandparent(mary, lisa).
5. ?- sibling(paul, susan).
6. Who are all the grandchildren of John?
?- grandparent(john, X).

7. Who are the ancestors of Emma?
?- ancestor(X, emma).
8. Find all siblings in the family.
?- sibling(X, Y).
9. Who are Emma's grandparents?
?- grandparent(X, emma).
10. Find all people who have at least one sibling.
?- sibling(X, _).
11. List all parent-child pairs.
?- parent(X, Y).
12. Who has both a mother and a father?
?- father(F, X), mother(M, X).